

- 2 As part of a physics lesson demonstration, a student stands between a sound speaker and a wall, such that he is 1.10 m from the speaker and 0.30 m from the wall as shown in Fig. 2.1. The speaker emits a sound wave of a single wavelength of 0.40 m. The sound wave hits the wall perpendicularly and is reflected back.

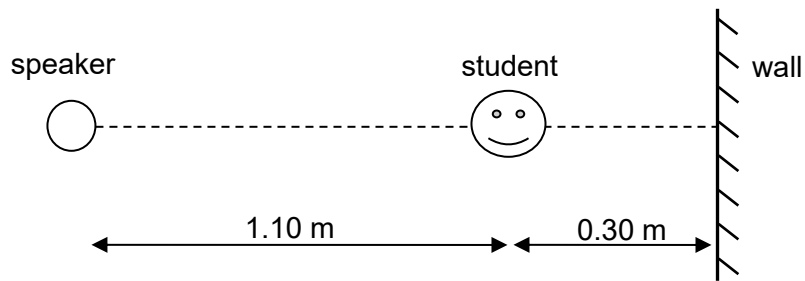


Fig. 2.1

The reflection at the wall causes the wave to undergo a phase change of π rad.

- (a) Calculate the phase difference between the waves that come directly from the speaker and the one that is reflected back from the wall, when they meet at the student's location. Leave your answer in multiples of π .

phase difference = rad [3]

- (b) The sound wave is emitted uniformly in all directions by the speaker. When the wave reaches the student directly from the speaker, it has an amplitude of 1.2×10^{-8} m.
- (i) Calculate the amplitude of the wave that reaches the student after reflection from the wall. State an assumption made in your calculation.

amplitude = m [2]

Assumption:

..... [1]

- (ii) Hence calculate the amplitude of the resultant wave experienced by the student.

amplitude = m [1]

- (c) The intensity of the sound wave that reaches the student directly from the speaker is $1.0 \times 10^{-6} \text{ W m}^{-2}$.

Calculate the intensity of the resultant wave that reaches him.

intensity = W m^{-2} [2]

3 Fig. 3.1 shows lines joining points at the same potential in and around a pair of parallel